**Algorithm 1** Dropout $(A_i, p)$

**Input:** Layer matrix $A_i \in \mathbb{R}^{h_i \times h_{i-1}}$, dropout probability $0 < p < 1$.
**Output:** Returns $\hat{A}_i \in \mathbb{R}^{h_i \times h_{i-1}}$.

1: For each $j \in [h_{i-1}]$, let $\delta_j$ be an i.i.d. Bernoulli random variable which takes the value 0 with probability $p$ and takes the value $\frac{1}{1-p}$ with probability $(1-p)$.

2: For each $j \in [h_{i-1}]$, let $[\hat{A}_i]_j$ be $\delta_j [A_i]_j$, where $[\hat{A}_i]_j$ and $[A_i]_j$ are the $j$-th column of $\hat{A}_i$ and $A_i$ respectively.

The main theorem that we prove in this section is:

**Theorem 2.** *Let $\theta^A$ and $\theta^B$ be two fully connected networks that are both $\epsilon$-noise stable, there exists a path with 10 line segments in parameter space $\pi : [0,1] \to \Theta$ between $\theta^A$ and $\theta^B$ such that*[4] $L(f_{\pi(t)}) \leq \max\{L(f_{\theta^A}), L(f_{\theta^B})\} + \widetilde{O}(\epsilon)$ *for* $0 \leq t \leq 1$.

To prove the theorem, we will first show that the networks $\theta^A$ and $\theta^B$ are $\widetilde{O}(\epsilon)$-dropout stable. This is captured in the following main lemma:

**Lemma 3.** *Let $\theta$ be an $\epsilon$-noise stable network, and let $\theta_1$ be the network with weight matrices from layer 2 to layer $d$ dropped out by Algorithm 1 with dropout probability $\widetilde{\Omega}(1/h_{min}) < p \leq \frac{3}{4}$. For any $2 \leq i \leq d$, assume $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ for $1 \leq j \leq h_{i-1}$. For any $0 \leq t \leq 1$, define the network on the segment from $\theta$ to $\theta_1$ as $\theta_t := \theta + t(\theta_1 - \theta)$. Then, with probability at least 1/4 over the weights generated by Algorithm 1, $L(f_{\theta_t}) \leq L(f_\theta) + \widetilde{O}(\sqrt{p}\epsilon)$, for any $0 \leq t \leq 1$.*

The main difference between Lemma 3 and Lemma 1 is that we can now directly interpolate between the original network and its dropout version, which reduces the number of segments required. This is mainly because in the noise stable setting, we can prove that after dropping out the neurons, not only does the output remains stable but moreover *every* intermediate layer also remains stable.

From Lemma 3, the proof of Theorem 2 is very similar to the proof of Theorem 1. The detailed proof is given in Section B.

The additional power of Lemma 3 also allows us to consider a smaller dropout probability. The theorem below allows us to trade the dropout fraction with the energy barrier $\epsilon$ that we can prove—if the network is highly overparametrized, one can choose a small dropout probability $p$ which allow the energy barrier $\epsilon$ to be smaller.

**Theorem 3.** *Suppose there exists a network $\theta^*$ with layer width $h_i^*$ for each layer $i$ that achieves loss $L(f_{\theta^*})$, and minimum hidden layer width $h_{min}^* = \widetilde{\Omega}(1)$. Let $\theta^A$ and $\theta^B$ be two $\epsilon$-noise stable networks. For any dropout probability $1.5 \max_{1 \leq i \leq d-1}(h_i^*/h_i) \leq p \leq 3/4$, if for any $2 \leq i \leq d$, $1 \leq j \leq h_{i-1}$, $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ then there exists a path with 13 line segments in parameter space $\pi : [0,1] \to \Theta$ between $\theta^A$ and $\theta^B$ such that $L(f_{\pi(t)}) \leq \max\{L(f_{\theta^A}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^B}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^*})\}$ for $0 \leq t \leq 1$.*

Intuitively, we prove this theorem by connecting $\theta^A$ and $\theta^B$ via the neural network $\theta^*$ with narrow hidden layers. The detailed proof is given in Section B.

# 5 Disconnected modes in two-layer nets

The mode connectivity property is not true for every neural network. Freeman and Bruna (2016) gave a counter-example showing that if the network is not overparametrized, then there can be different global minima of the neural network that are not connected. Venturi et al. (2018) showed that spurious valleys can exist for 2-layer ReLU nets with an arbitrary number of hidden units, but again they do not extend their result

columns in $A_1$ would drop out input coordinates, which is not necessary.

[4] Here $\widetilde{O}(\cdot)$ hides log factors on relevant factors including $|S|, d, \|x\|, 1/\epsilon$ and $h_i\|A_i\|$ for layers $i \in [d]$.

to the overparametrized setting. In this section, we show that even if a neural network is overparametrized—in the sense that there exists a network of smaller width that can achieve optimal loss—there can still be two global minimizers that are not connected.

In particular, suppose we are training a two-layer ReLU student network with $h$ hidden units to fit a dataset generated by a ground truth two-layer ReLU teacher network with $h_t$ hidden units such that the samples in the dataset are drawn from some input distribution and the labels computed via forward passes through the teacher network. The following theorem demonstrates that regardless of the degree to which the student network is overparametrized, we can always construct such a dataset for which global minima are not connected.

**Theorem 4.** *For any width $h$ and and convex loss function $l : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$ such that $l(y, \hat{y})$ is minimized when $y = \hat{y}$, there exists a dataset generated by ground-truth teacher network with two hidden units (i.e. $h_t = 2$) and one output unit such that global minimizers are not connected for a student network with $h$ hidden units.*

Our proof is based on an explicit construction. The detailed construction is given in Section C.

# 6 Experiments

We now demonstrate that our assumptions and theoretical findings accurately characterize mode connectivity in practical settings. In particular, we empirically validate our claims using standard convolutional architectures—for which we treat individual filters as the hidden units and apply channel-wise dropout (see Remark 1)—trained on datasets such as CIFAR-10 and MNIST.

Training with dropout is not necessary for a network to be either dropout-stable or noise-stable. Recall that our definition of dropout-stability merely requires the existence of a particular sub-network with half the width of the original that achieves low loss. Moreover, as Theorem 3 suggests, if there exists a narrow network that achieves low loss, then we need only be able to drop out a number of filters equal to the width of the narrow network to connect local minima.

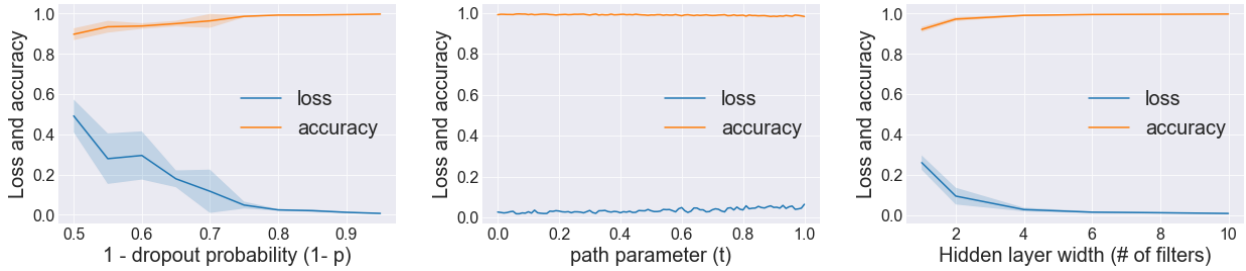


Figure 2: Results for convolutional networks trained on MNIST.

First, we demonstrate in the left plot in Figure 2 on MNIST that 3-layer convolutional nets (not counting the output layer) with 32 $3 \times 3$ filters in each layer tend to be fairly dropout stable—both in the original sense of Definition 1 and especially if we relax the definition to allow for wider subnetworks—despite the fact that no dropout was applied in training. For each trial, we randomly sampled 20 dropout networks with *exactly* $\lfloor 32(1-p) \rfloor$ non-zero filters in each layer and report the performance of the best one. In the center plot, we verify for $p = 0.2$ we can construct a linear path $\pi(t) : \mathbb{R} \to \Theta$ from our convolutional net to a dropout version of itself. Similar results were observed when varying $p$. Finally, in the right plot we demonstrate the existence of 3-layer convolutional nets just a few filters wide that are able to achieve low loss on MNIST. Taken together, these results indicate that our path construction in Theorem 3 performs well in practical settings. In particular, we can connect two convolutional nets trained on MNIST by way of first interpolating between the original nets and their dropped out versions with $p = 0.2$, and then connecting the dropped out versions by way of a narrow subnetwork with at most $\lfloor 32p \rfloor$ non-zero filters.

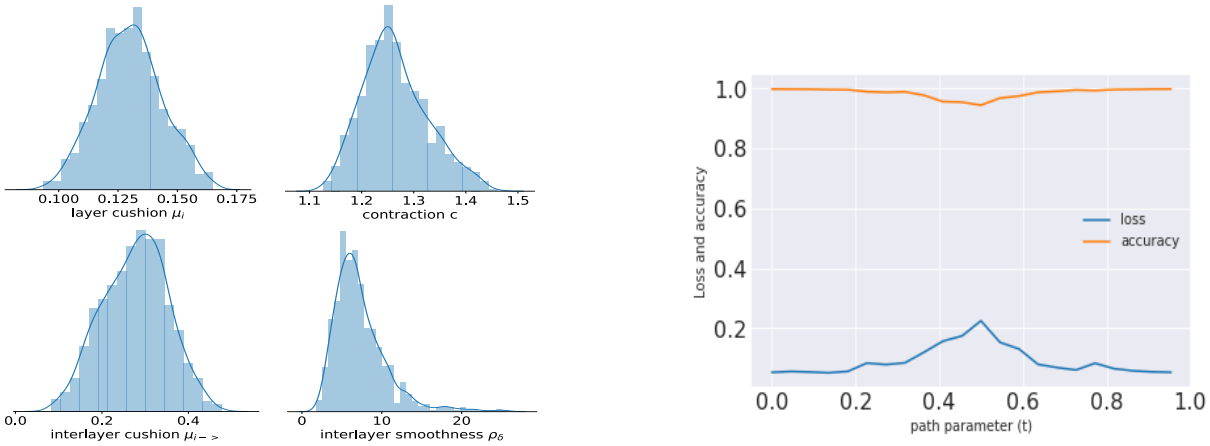


Figure 3: Left) Distribution of layer cushion, activation contraction, interlayer cushion and interlayer smoothness of the 6-th layer of a VGG-11 network on the training set. The other layers' parameters are exhibited in Section D.3. Right) The loss and training accuracy along the path between two noise stable VGG-11 networks described in Theorem 3.

We also demonstrate that the VGG-11 (Simonyan and Zisserman, 2014) architecture trained with channel-wise dropout (Tompson et al., 2015; Keshari et al., 2018) with $p = 0.25$ at the first three layers[5] and $p = 0.5$ at the others on CIFAR-10 converges to a noise stable minima—as measured by layer cushion, interlayer cushion, activation contraction and interlayer smoothness. The network under investigation achieves 95% training and 91% test accuracy with channel-wise dropout *activated*, in comparison to 99% training and 92% test accuracy with dropout turned off. Figure 3 plots the distribution of the noise stability parameters over different data points in the training set, from which we can see they behave nicely. Interestingly, we also discovered that networks trained without channel-wise dropout exhibit similarly nice behavior on all but the first few layers. Finally, in Figure 3, we demonstrate that the training loss and accuracy obtained via the path construction in Theorem 3 between two noise stable VGG-11 networks $\theta_A$ and $\theta_B$ remain fairly low and high respectively—particularly in comparison to directly interpolating between the two networks, which incurs loss as high as 2.34 and accuracy as low as 10%, as shown in Section D.2.

Further details on all experiments are provided in Section D.1.

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[5]we find the first three layers are less resistant to channel-wise dropout.